

Name: _____

Date: _____

Standards: MA.5.AR.1.1; MA.5.FR.1.1; MA.5.NSO.1.1-1.5

Show your work in the box for each problem.

Write your final answer on the answer line (or in the blanks shown).

Use the words and strategies from Unit 15: percentage.

1 Recall: Meaning of Percent

What does 'percent' mean? Complete: 47% means ____ out of ____.

Fill in the blanks.

Percent means 'per hundred.' 47% means 47 out of 100. _____

2 Recall: ConversionsConvert each: 0.65 to a percentage; $\frac{3}{4}$ to a percentage; 90% to a decimal; 30% to a fraction.

Show your work for each.

0.65 = 65%; $\frac{3}{4}$ = 75%; 90% = 0.9; 30% = $\frac{3}{10}$. _____**3 Application: Percentage of a Number**

Find 45% of 160.

Show your work.

 $0.45 \times 160 = 72$. _____

4 Application: Percentage Score

A student answered 36 out of 48 questions correctly.

What percentage did she get correct?

$$36/48 = 3/4 = 75\%. \quad \underline{\hspace{2cm}}$$

5 Word Problem: Discount

A \$150 video game console is on sale for 30% off.

What is the sale price?

$$\text{Discount} = 0.30 \times \$150 = \$45. \text{ Sale price} = \$150 - \$45 = \$105. \quad \underline{\hspace{2cm}}$$

6 Word Problem: Survey

In a survey of 200 people, 35% said they exercise daily.

How many exercise daily? How many do not?

$$0.35 \times 200 = 70 \text{ exercise daily. } 200 - 70 = 130 \text{ do not. } \underline{\hspace{2cm}}$$

7 Word Problem: Tip

A restaurant bill is \$85. Leo wants to leave an 18% tip.

How much is the tip? What is the total amount paid?

Tip = $0.18 \times \$85 = \15.30 . Total = $\$85 + \$15.30 = \$100.30$. _____

8 Find the Mistake

A student converts $\frac{2}{5}$ to a percentage and writes 25%.

Find the error and show the correct conversion.

$\frac{2}{5} = \frac{40}{100} = 40\%$. The student confused $\frac{2}{5}$ with $\frac{1}{4}$ (which is 25%). Correct answer: 40%. _____

9 Multi-Step: Tax and Stay

A hotel room costs \$120 per night. There is an 8% tax and the guest stays 3 nights.

What is the total cost for 3 nights including tax?

3 nights = \$360. Tax = $0.08 \times \$360 = \28.80 . Total = \$388.80. _____

10 Reasoning: Percentage Increase

Mira had 40 followers on a blog. After one month, she has 52 followers.

By what percentage did her followers increase? Show your work.

Increase = 12. Percentage increase = $(12 \div 40) \times 100 = 30\%$. _____